

ProspectMap: The Two Ways a Mineral Prospectivity Map Lies, Quantified on the Real US Mississippi-Valley-Type Belt

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Not peer reviewed. Source code, data and computational records: github.com/fsantibanezleal/CAOS_ProspectMap

Abstract. A mineral prospectivity map stacks geophysical, geochemical and structural evidence over a study area and outputs, per cell, the probability that a deposit is there. Such maps are typically presented with an area-under-curve (AUC) and a posterior, and both numbers can be inflated in two specific, well-understood ways. ProspectMap is an interactive tool that computes such a map by Weights of Evidence, a Bayesian log-odds update, and makes those two inflations *first-class*, measuring them rather than hiding them. This report quantifies both on the real US Midcontinent Mississippi-Valley-type zinc-lead belt (the open Critical Minerals Mapping Initiative dataset of Lawley *et al.*, 2022) and on ten synthetic cases. The first inflation is *evaluation*: reporting AUC under random cross-validation, which lets spatially adjacent cells leak between train and test, overstates prospective skill. On the real belt, with thresholds, weights and prior fitted on the training folds only and every map cell scored, the Weights-of-Evidence AUC falls from 0.74 under random cross-validation to 0.65 under spatial block cross-validation, an inflation of 0.09, the second largest of the eleven cases. Under the same spatial folds a learned classifier reaches 0.91, but a baseline that ranks cells by their distance to the nearest training deposit reaches 0.90, so this protocol does not show that the classifier learned more than proximity to known deposits. The second inflation is *modelling*: Weights of Evidence assumes the evidence layers are conditionally independent given a deposit, and on the real belt they are not. The posterior probabilities sum to 1124 against 858 deposit cells (omnibus $z = 8.3$), so the posterior is overconfident. The six layers' contrasts are individually significant (studentized contrasts 2.45 to 5.80; exact one-sided $p \leq 1.9 \times 10^{-3}$), but four of them come from thresholds that leave no deposit outside the pattern, where the weight outside the pattern is set by a zero-count correction. The tool displays the spatial-CV AUC, the conditional-independence test and the per-layer significance beside the inflated values; every number and figure can be regenerated from the released artifacts.

Index Terms. Mineral prospectivity, Weights of Evidence, conditional independence, spatial cross-validation, AUC inflation, Mississippi-Valley-type, critical minerals, model evaluation, reproducibility.

I. INTRODUCTION

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Interactive tool and reproducible pipeline (MIT) at https://github.com/fsantibanezleal/CAOS_ProspectMap; the tool is available at <https://prospectmap.fasl-work.com>.

WHERE is the next ore deposit most likely to be? Mineral prospectivity mapping answers it by overlaying evidence layers, a magnetic high here, a geochemical anomaly there, proximity to a favourable structure, over a study-area grid and computing, per cell, a posterior probability of a deposit. The classical method is Weights of Evidence [2], [1]: each binary evidence pattern gets a positive weight W^+ where it is present and W^- where it is absent, their difference is the contrast $C = W^+ - W^-$, and the posterior log-odds of a deposit in a cell is the prior log-odds plus the sum of the present-or-absent weights. The output is a map, and the map is usually reported with two numbers: a high area-under-curve showing it separates known deposits from barren ground, and confident-looking weights. Both are, in a well-documented way, prone to inflation.

ProspectMap is a tool built to expose the inflation. It computes the Weights-of-Evidence posterior live, and it makes first-class the two ways the numbers lie. The first is that an AUC reported under *random* cross-validation is optimistic, because a prospectivity grid is spatially autocorrelated and a random split leaves near-neighbour cells in both train and test, so the model is partly tested on what it has seen; *spatial* cross-validation, holding out contiguous blocks, gives a realistic estimate [5]. The second is that Weights of Evidence assumes the evidence layers are conditionally independent given a deposit, and when they are not the posterior is overconfident [3]; and a contrast can look large because of how its binarization threshold was chosen. This report quantifies both, on the real US Mississippi-Valley-type zinc-lead belt from the open Critical Minerals Mapping Initiative dataset [4] and on ten synthetic cases (four terranes, two data regimes and four controls), and reports a learned classifier [6] and a distance-to-deposit baseline under the same folds. The contribution is the measurement and the display of the two inflations, not a new prospectivity algorithm.

II. THE FIRST LIE: CROSS-VALIDATION INFLATION

The first inflation is a matter of how the map is scored, and it is quantified in Fig. 1a. For every case, the Weights-of-Evidence posterior is evaluated two ways: random five-fold cross-validation, which splits the grid cells at random, and

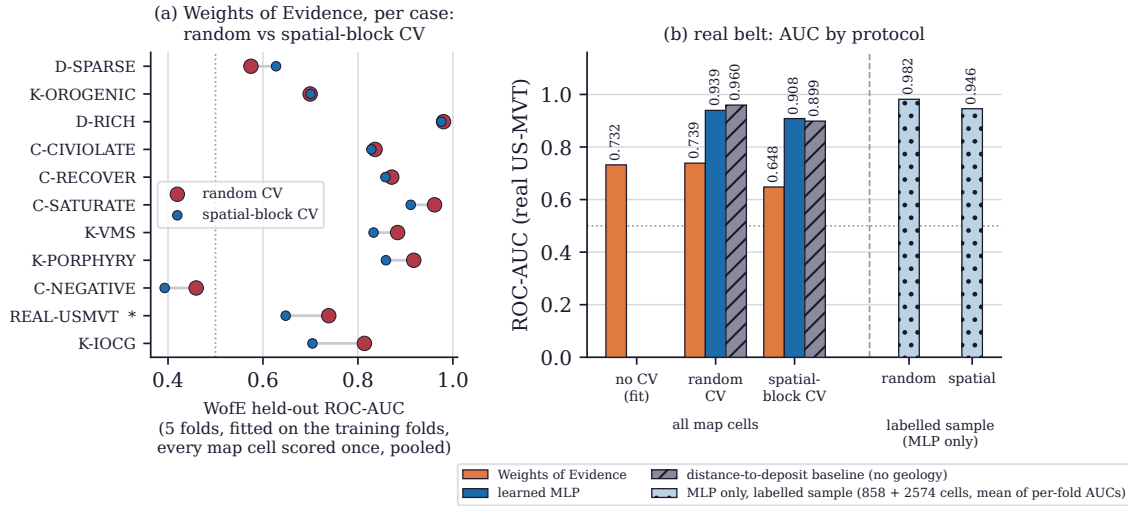


Fig. 1. The evaluation inflation. (a) Weights-of-Evidence AUC under random versus spatial cross-validation, per case (five folds; spatial folds are 20×20 -cell blocks; thresholds, weights and prior fitted on the training folds’ cells; every map cell scored once, pooled): random cross-validation (red) is at least as high as spatial (blue) in every case except K-OROGENIC and D-SPARSE, where spatial exceeds random by 0.001 and 0.053; the gap is largest for the synthetic K-IOCG terrane (0.110) and second largest on the real US-MVT belt (marked *), where the AUC falls from 0.739 to 0.648. (b) The real belt, AUC by protocol: Weights of Evidence without cross-validation (0.732, fitted on all deposits); over all map cells with the same folds, random and spatial cross-validation of Weights of Evidence (0.739, 0.648), the learned classifier (0.939, 0.908) and a baseline that ranks cells by their distance to the nearest training deposit (0.960, 0.899); set apart, the classifier on its labelled sample (858 deposit cells and 2574 non-deposit cells more than two cells from any deposit; mean of per-fold AUCs; 0.982, 0.946).

spatial five-fold cross-validation, which holds out contiguous 20×20 -cell blocks so that the test set is geographically separate from the training set. In both, each layer’s binarization threshold, its weights and the prior are fitted on the training folds’ cells only, the held-out fold is only scored, and every map cell receives one held-out score. Random cross-validation is at least as optimistic in every case except K-OROGENIC and D-SPARSE, where the spatial AUC is higher by 0.001 and 0.053, because a prospectivity grid is strongly spatially autocorrelated and a random split leaves a test cell’s near neighbours in the training set, so the model is scored partly on cells like ones it has seen. On the *real* US Mississippi-Valley-type belt the gap is the second largest of the eleven cases, after the synthetic K-IOCG terrane (0.110): the Weights-of-Evidence AUC falls from 0.739 under random cross-validation to 0.648 under spatial, an inflation of 0.091. The spatial estimate is the relevant one; the random one overstates how well the map would find a deposit in unseen ground.

Fig. 1b sets these values beside the Weights-of-Evidence AUC without cross-validation (0.732, weights fitted and scored on the whole map) and beside two other models scored on every map cell with the same folds. A learned classifier, a small multilayer perceptron on the six continuous evidence layers (Weights of Evidence uses their binarized patterns), reaches 0.939 under random and 0.908 under spatial cross-validation. A baseline that learns no geology and ranks each cell by its distance to the nearest training deposit reaches 0.960 and 0.899: the classifier exceeds it by 0.010 under spatial folds and falls below it under random ones. Spatial blocks of 20×20 cells still leave each held-out block beside training blocks, so this protocol cannot separate what the classifier learned from the evidence layers from proximity to known deposits.

A logistic regression on the same binary patterns, also fitted out of fold, tracks Weights of Evidence (0.744 random, 0.644 spatial). On its own labelled sample, the 858 deposit cells and 2574 sampled non-deposit cells more than two cells from any deposit, scored as the mean of per-fold AUCs, the classifier gives 0.982 and 0.946; those values come from a different cell set and are not comparable with the all-cells ones.

III. THE SECOND LIE: AN OVERCONFIDENT POSTERIOR

The second inflation is in the model. Weights of Evidence assumes the evidence layers are conditionally independent given a deposit; when correlated layers are stacked, their weights double-count the same signal and the posterior becomes overconfident, which is why a conditional-independence test [3] belongs in any Weights-of-Evidence workflow. On the real belt the assumption fails. Under conditional independence the posterior probabilities summed over the map equal, in expectation, the number of deposit cells; here they sum to $T = 1124$ against $N(D) = 858$, with $s(T) = 32$, so the omnibus test gives $z = 8.3$ [3] and the conditional-independence ratio $N(D)/T$ is 0.76. The posterior probabilities therefore overstate the number of deposits by 31%. The pairwise tests, stratified by deposit presence, locate the dependence: the *lab* and *faultprox* patterns are the most dependent pair (Cramér’s $V = 0.79$), followed by *grav* with *faultprox* (0.31) and *grav* with *lab* (0.28).

The individual weights are a separate question (Fig. 2). All six contrasts are statistically significant: their studentized contrasts $C/s(C)$ range from 2.45 (*mag*) to 5.80 (*marginprox*), and exact one-sided Fisher tests on the six 2×2 tables give $p \leq 1.9 \times 10^{-3}$ for every layer. Their sizes, however, reflect how the thresholds were chosen. Each layer is binarized

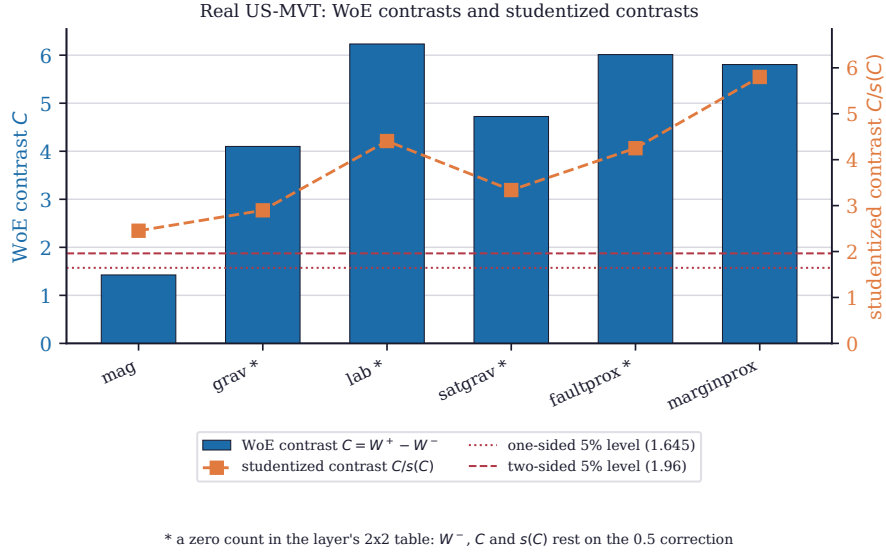


Fig. 2. The weights of the real US-MVT belt. Weights-of-Evidence contrasts $C = W^+ - W^-$ (bars) and studentized contrasts $C/s(C)$ (line) of the six evidence layers at the thresholds chosen on all deposits, with the one-sided (1.645) and two-sided (1.96) 5% levels. Every layer exceeds both. The four starred layers have no deposit cell outside their pattern, so their W^- , C and $s(C)$ rest on the 0.5 zero-count correction.

at the threshold that maximizes C among 40 candidates with $C/s(C) \geq 1.5$, and for `grav`, `lab`, `satgrav` and `faultprox` that threshold puts every deposit cell inside the pattern. W^- then rests on a zero count and on the correction of 0.5 that replaces it; with a correction of 0.1 the same four layers give $C/s(C)$ between 1.81 and 2.48. The exact tests do not depend on the correction, but the magnitudes of C , up to 6.2, do, and they are maxima over 40 thresholds on the same data, with $s(C)$ computed as if the 25,344 cells were independent, although the layers and the deposits are spatially autocorrelated. The weights are significant, but not as decisive as their raw contrasts suggest.

IV. DISCUSSION AND SCOPE

The two inflations are independent and both matter. A map can pass the conditional-independence test and still report an inflated AUC because it was scored with random cross-validation; or it can report a spatially cross-validated AUC and still carry an overconfident posterior. ProspectMap's contribution is to compute the Weights-of-Evidence map and then put the deflating diagnostics next to it: the spatial-CV AUC beside the random one, the conditional-independence test beside the posterior, and the studentized contrast, flagged when it rests on a zero-count correction, beside the raw one. A learned classifier and a distance-to-deposit baseline are reported under the same folds; on the real belt the classifier's spatial-CV AUC (0.908) is far above that of Weights of Evidence (0.648), but the distance baseline reaches 0.899, so the comparison ranks proximity to known deposits as much as geology. The ten synthetic cases, four terranes, two data regimes and four controls, exercise the diagnostics under known ground truth; the one real case is the open Critical Minerals Mapping Initiative US-MVT dataset [4], on which spatial cross-validation removes the second-largest AUC inflation of the eleven cases and the

conditional-independence test fails with $z = 8.3$. The tool is a decision-support and teaching instrument, not an exploration recommendation for any specific ground; it says where the evidence points and, more importantly, how much to trust it. The interface shows the same diagnostics on every map.

V. CONCLUSION

A mineral prospectivity map lies in two well-understood ways: it reports an AUC inflated by random cross-validation, and its Weights-of-Evidence posterior is overconfident when the evidence layers are not conditionally independent given a deposit. ProspectMap measures both. On the real US Mississippi-Valley-type belt the Weights-of-Evidence AUC falls from 0.74 to 0.65 when scored by spatial instead of random cross-validation, with thresholds, weights and prior fitted on the training folds only; a learned classifier reaches 0.91 under the same spatial folds, where a baseline that ranks cells by their distance to the nearest training deposit reaches 0.90; and the posterior probabilities sum to 1.31 times the number of deposit cells (conditional-independence $z = 8.3$). The six layers' contrasts are individually significant, although four rest on thresholds that leave no deposit outside the pattern. The contribution is a reproducible tool that displays these diagnostics next to the inflated values.

APPENDIX A REPRODUCTION

The repository artifacts under `data/derived/` carry each case's Weights-of-Evidence layers (counts, contrasts and studentized contrasts), posterior summary, pairwise and omnibus conditional-independence tests, and both random- and spatial-cross-validation AUC (`case-results.json`). `pm-learned-real.json` carries the real belt's comparison under the same folds (`spatial_cv` and

random_cv: Weights of Evidence, the learned classifier, the logistic regression and the distance baseline), the Weights-of-Evidence AUC without cross-validation (nocv) and the classifier's labelled-sample values (labelled_sample_cv); the real US-MVT trace (REAL-USMVT) carries the Lawley *et al.* provenance. manuscripts/prospectivity/data/build_pm.py rebuilds the manuscript data (data/pm.json) from those artifacts, including the exact Fisher tests, and figures/make_figs.py draws Fig. 1 and Fig. 2 from it; the pipeline is seeded.

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